

P S S Krishna Alapati

Data Analyst | BI Developer | SQL & Python Analytics | AI Driven Decisions

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PROFESSIONAL SUMMARY

- Data Analyst with 3+ years of experience cleaning, analyzing, and visualizing large-scale datasets using Power BI, Tableau, SQL, and Python to deliver KPI-driven insights across research and business environments.
- Built end-to-end ETL pipelines and interactive dashboards (Power BI, Tableau, DAX, Power Query) translating complex data into executive-ready reporting for stakeholder decision-making.
- Skilled in advanced SQL — CTEs, stored procedures, window functions, indexing, execution-plan tuning — across SQL Server, PostgreSQL, MySQL, Snowflake, and Amazon Redshift.
- Experienced in statistical analysis, forecasting, A/B testing, predictive analytics, and data governance (RBAC, RLS, data quality frameworks, metadata management).
- Deployed cloud analytics solutions on Microsoft Azure, AWS (S3, Redshift, EC2, RDS), and Databricks, integrating structured and semi-structured sources into unified reporting models.
- AWS Certified Solutions Architect with hands-on dimensional modeling (star/snowflake schema), OLAP cube design, and data warehousing best practices.
- Optimized dashboard query performance, indexing, and incremental refresh for large-scale datasets.

TECHNICAL SKILLS

- **BI & Visualization:** Power BI (Desktop & Service, DAX, Power Query, RLS, Incremental Refresh), Tableau (Desktop & Server, LOD, Parameters), Power Apps, Power Automate, or Copilot.
- **Programming Languages:** SQL, Python (Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, Seaborn), Bash / Shell Scripting, JavaScript, C++, JAVA
- **Databases:** Microsoft SQL Server (SSMS), PostgreSQL, MySQL, Amazon Redshift, Snowflake, Oracle, NoSQL
- **Data Engineering & ETL:** SSIS, Apache Airflow, PySpark, Delta Lake, Power Query, ETL Design & Development
- **Data Warehousing & Modeling:** Dimensional Modeling, Star/Snowflake Schema, Data Marts, OLTP, OLAP, Tabular Models, Cubes, SSAS
- **Cloud Platforms:** Microsoft Azure, AWS (S3, Redshift, EC2, RDS), Databricks
- **Analytics:** KPI Development, Data Analysis, A/B Testing, Hypothesis Testing, Regression Analysis, Forecasting, Predictive Analytics, EDA, Anomaly Detection, Feature Engineering, Using AI for efficient analysis, Query Writing Using AI, Data Cleaning, Data Collection, Data Profiling, Data Wrangling, Data manipulation, Data integrity, Tracking Operational metrics, Data Management, Data Reporting
- **Governance & Compliance:** Data Quality Frameworks, Data Validation, Data Mapping, RBAC, RLS, Audit Reporting, Data Governance, Metadata Management
- **Tools & Methods:** Advanced Microsoft Excel (Pivot Tables, VLOOKUP/XLOOKUP), JIRA, Confluence, Git, GitHub, MS Teams, SAP, ERP, Agile / Scrum, Waterfall, SDLC, MDM

EDUCATION

- **University of Alabama at Birmingham** *Birmingham, Alabama*
 - Master of Science in Computer Science | GPA: 3.4/4.0 | January 2024 – December 2025
 - **Relevant Coursework:** Foundations of Data Science, Machine Learning, Advanced Algorithms & Applications, Cloud Security, Network Security, Computer Security, Database Systems, Computer Vision
- **KKR & KSR Institute of Technology and Sciences** *Guntur, India*
 - Bachelor of Technology in Computer Science | GPA: 7.35/10 | April 2019 – August 2023

PROFESSIONAL EXPERIENCE

University of Alabama at Birmingham – NVL Lab, AL

Data Analyst

Dec 2024 – March 2026

Responsibilities:

- Cleaned and transformed 95% of raw **HoloBMI neuroscience datasets** using Python (Pandas) and custom validation pipelines, achieving 98% data accuracy across **multi-gigabyte** experimental files.

- Synchronized microscope, high-speed camera, and behavioral event log timestamps across heterogeneous data streams, reaching 99% alignment accuracy and ensuring reliable downstream experimental analysis.
- Automated repetitive data processing workflows using **SLURM-based batch scheduling**, cutting manual pipeline execution time by 60% and freeing researcher bandwidth for active analysis.
- Engineered **ETL workflows** that ingested, mapped, and validated data from multiple experimental sources and Assisted in **data collection, cleaning**, and validation from various sources — **applying transformation logic, referential checks**, and schema enforcement before loading into analytical layers.
- Designed and implemented star-schema dimensional data models to support scalable **OLAP-style reporting, KPI computation**, and **performance monitoring dashboards for lab researchers**.
- Developed comprehensive SQL unit test suites for complex data transformations and query logic, **ensuring data quality**, integrity, and compliance with enterprise reporting standards.
- Built dynamic **Power BI** dashboards that visualized behavioral insights, experiment trends, and sensor data — enabling researchers to **identify anomalies** and make data-driven decisions in near real time.
- Optimized **Power BI data extracts** and underlying **SQL** queries to sustain responsive performance across large-scale multi-session reporting environments.
- Enforced **data governance** and **access security** by configuring Tableau Server permissions **with RBAC and RLS** policies, maintaining data privacy and compliance across research teams.
- Performed data profiling and **root cause analysis** to resolve inconsistencies, improving dataset reliability for publication-grade reporting.
- Used Advanced **Excel** (Pivot Tables, XLOOKUP, array formulas) for pre-dashboard data modeling and quality checks.

Environment / Tech Stack:

Python (Pandas, NumPy), SQL (CTEs, Stored Procedures, Unit Testing), Power BI (Desktop & Service, DAX), Tableau Server (RBAC, RLS), SLURM Batch Processing, ETL Pipeline Design, Dimensional Modeling (Star Schema), OLAP, Advanced Excel, Data Governance, Git

Cyient, India

Data Analyst

Feb 2022 – Dec 2023

Responsibilities:

- Developed and published interactive Power BI dashboards with **DAX measures, KPI cards, trend analysis**, and drill-through visuals to track product performance across multiple business units — reducing manual reporting effort by 25%.
- Integrated data from SQL Server, **AWS S3, Excel, JSON/XML** feeds, and Databricks into unified semantic models using Power Query, eliminating siloed reporting and reducing manual data consolidation by 40%.
- Engineered ETL pipelines to extract and transform data from **NoSQL** and relational sources, applying schema mapping, data type enforcement, and **aggregation logic** before loading into analytical platforms.
- Wrote and optimized SQL queries — including **CTEs, window functions**, and indexed views — improving query execution time by 15% and reducing dashboard load latency across high-traffic reporting environments.
- Designed **A/B testing** frameworks in **Tableau using hypothesis testing**, Bayesian analysis, and causal inference, improving campaign performance by 15%.
- Automated data cleansing, transformation, and validation workflows using **Python (Pandas, NumPy)**, replacing manual preprocessing steps with repeatable, auditable pipelines.
- Conducted predictive analytics using **Scikit-learn regression and classification models** on structured datasets to support **sales forecasting, trend projection**, and operational planning decisions.
- Performed structured **exploratory data analysis (EDA)** on both structured and semi-structured datasets — identifying patterns, outliers, and seasonal trends to inform stakeholder reporting.
- Configured **role-based access control (RBAC)** and row-level security (RLS) in **Power BI**, implementing burst reporting schedules to ensure secure, role-specific report delivery.
- Established centralized BI governance standards — including KPI definitions, version control, data dictionaries, and **metadata documentation** — ensuring reporting consistency across product, marketing, and engineering teams.
- Assisted in designing dimensional data models using **star and snowflake schemas** to support analytical reporting scalability and warehouse query performance.
- Used Excel (VLOOKUP/XLOOKUP, Pivot Tables, advanced formulas) for quick data analysis and reporting tasks.

- Generated ad-hoc analytical reports and conducted **root cause analysis** on operational anomalies, delivering targeted insights that supported performance improvement decisions.

Environment:

Power BI, Tableau, SQL Server, AWS, Databricks, Apache Airflow, Python (Pandas, NumPy, Scikit-learn), Power Query, Excel (VLOOKUP/XLOOKUP, Pivot Tables), Git, NoSQL databases, ETL, JSON/XML, Data Modeling (Star/Snowflake Schema), A/B Testing & Statistical Analysis, EDA

PROJECTS

SQL Data Exploration & Power BI Dashboard

- Engineered end-to-end ETL pipelines using Power Query to connect Power BI to MySQL and Excel sources, applying data cleansing, schema validation, and transformation logic.
- Developed advanced SQL queries, stored procedures, and statistical aggregations to calculate core business metrics — Profit & Loss, Net Error, Forecast Accuracy.
- Designed interactive Power BI dashboards with DAX measures, KPI indicators, and trend analysis for executive-level reporting.
- Executed UAT, resolved performance bottlenecks, and configured Power BI Gateway for automated scheduled refresh.

Loan Default Risk Analysis – Power BI & Snowflake

- Analyzed a financial loan dataset in Snowflake spanning demographics, credit scores, DTI ratios, and employment type to identify default patterns.
- Designed a star-schema data model and ETL pipelines supporting scalable, high-performance financial risk analysis in a cloud warehouse.
- Developed DAX measures for default rates, YoY loan volume changes, and credit-score performance bands.
- Built Power BI dashboards highlighting default behavior by borrower segment and credit category, supporting improved lending decisions.

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Solutions Architect – Associate:** Validated expertise in designing cloud-based data architectures using Amazon S3, Redshift, EC2, and RDS.
- IBM Data Analysis Professional Certificate:** Completed professional certification covering data wrangling, visualization, statistical analysis, and Python-based analytics workflows.
- Published a paper entitled **Web Application for Online Crop Marketing in IJMRSET** (Monthly, Peer Reviewed Referred Journal) e-ISSN: **2582-7219**
- Secured 2nd place (out of 100 teams) at the KITS AI/ML Hackathon — engineered a functional Decentralized Video Streaming Platform prototype as team lead of a 6-member group in a 48-hour competition.
- Active member and coordinator of UAB AIS, contributing to analytics workshops, student engagement events, and knowledge-sharing initiatives.
- Active member and coordinator of **INFOMAC** (college programming club) — organized coding bootcamps and technical seminars to build community technical skills.